

A Data-Calibrated ODE Model of MAPK & AKT Signaling Identifies Robust, Schedule-Dependent Strategies for Controlling Tumor Burden in BRAF-Mutant Melanoma

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01 Background

Melanoma is a MAPK-driven disease

- 5th most common cancer worldwide; over 40% of skin-cancer deaths (*Didier 2024*)
- ~50 to 60% carry **BRAF V600E**, hyper-activating MAPK (BRAF to MEK to ERK) and driving uncontrolled proliferation (*Davies 2002*)

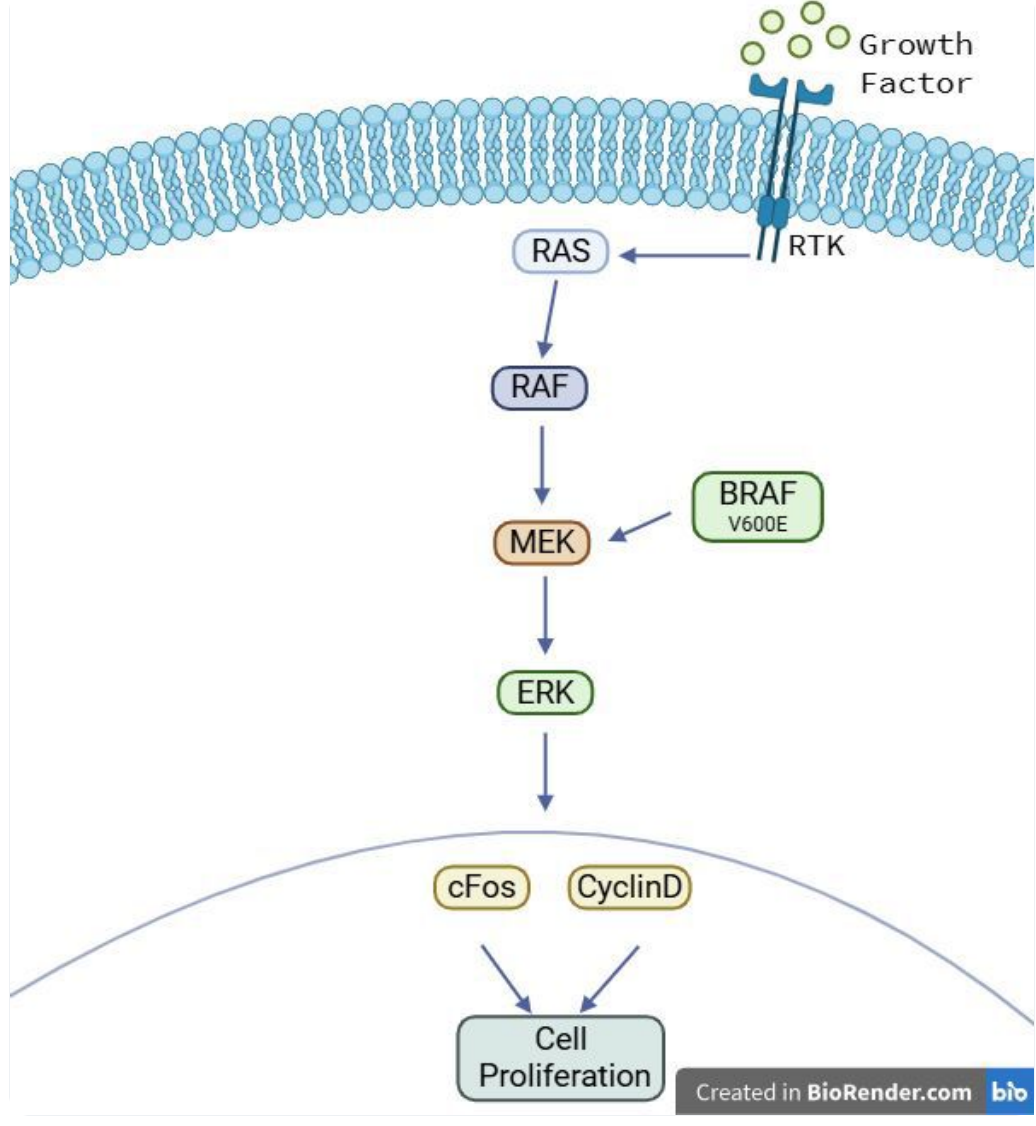
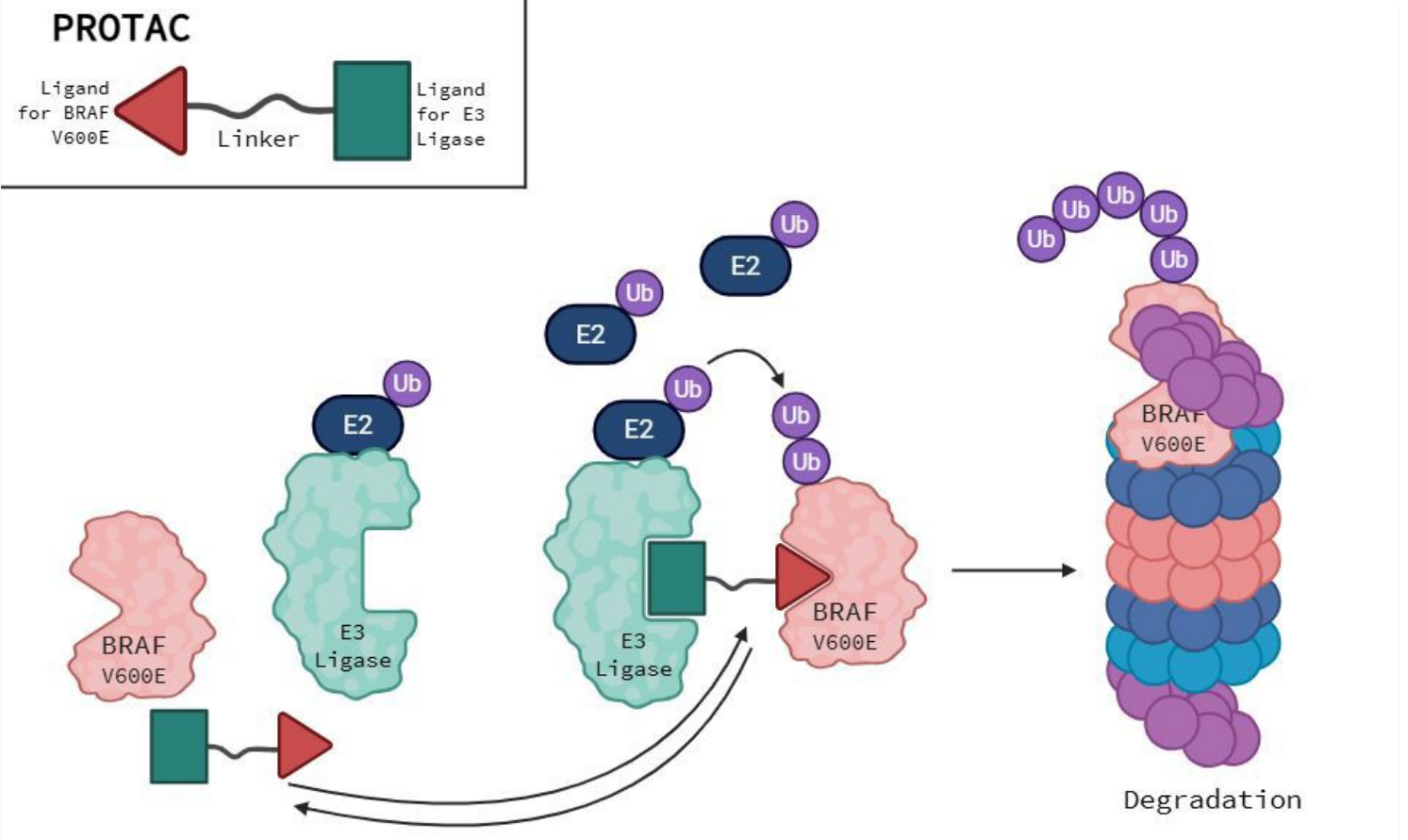


Fig 1 · MAPK pathway with the BRAF V600E mutation. BioRender.

Degrading BRAF with PROTACs



Created in BioRender.com bio

Fig 2 · PROTAC-mediated ubiquitination and degradation of mutant BRAF. BioRender.

- A bifunctional linker joins a BRAF-V600E ligand to an E3-ligase ligand, driving ubiquitination and proteasomal degradation
- Unlike occupancy inhibitors it removes the protein entirely; catalytic action gives deeper MAPK blockade (*Alabi 2021*, over 95% degradation)

Resistance is inevitable

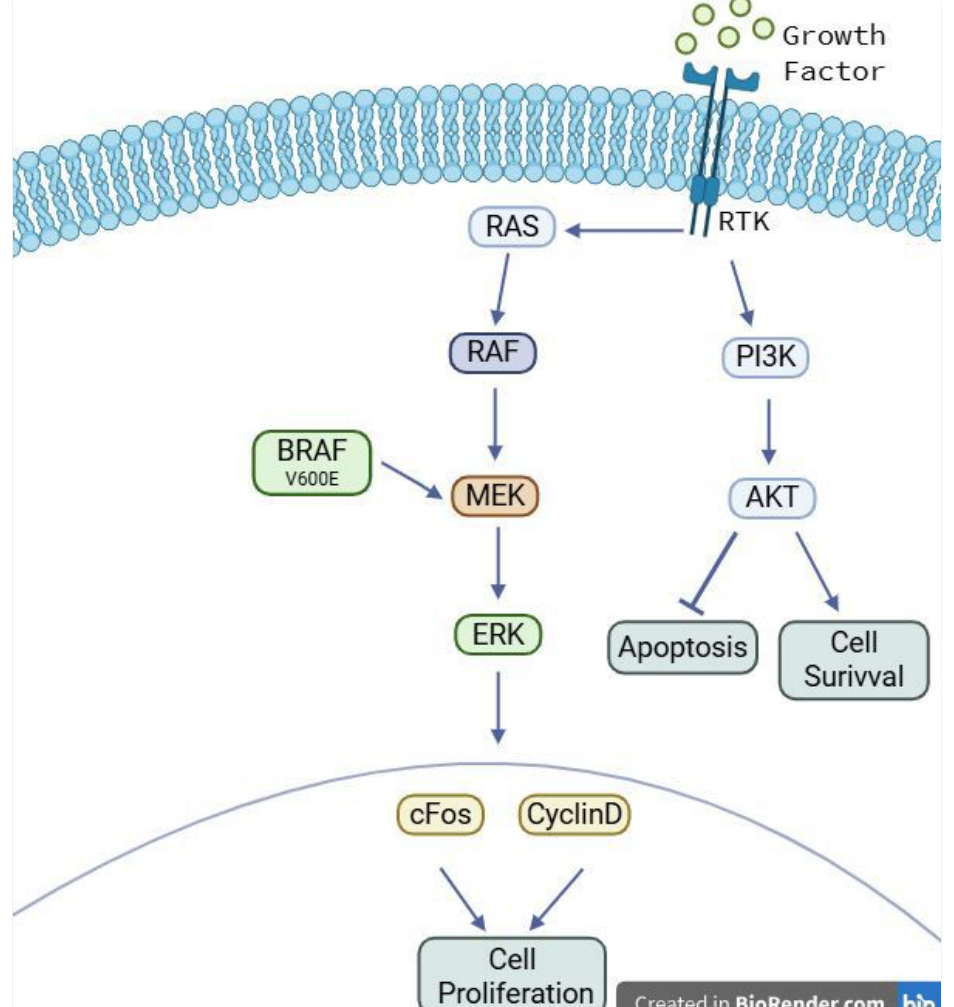
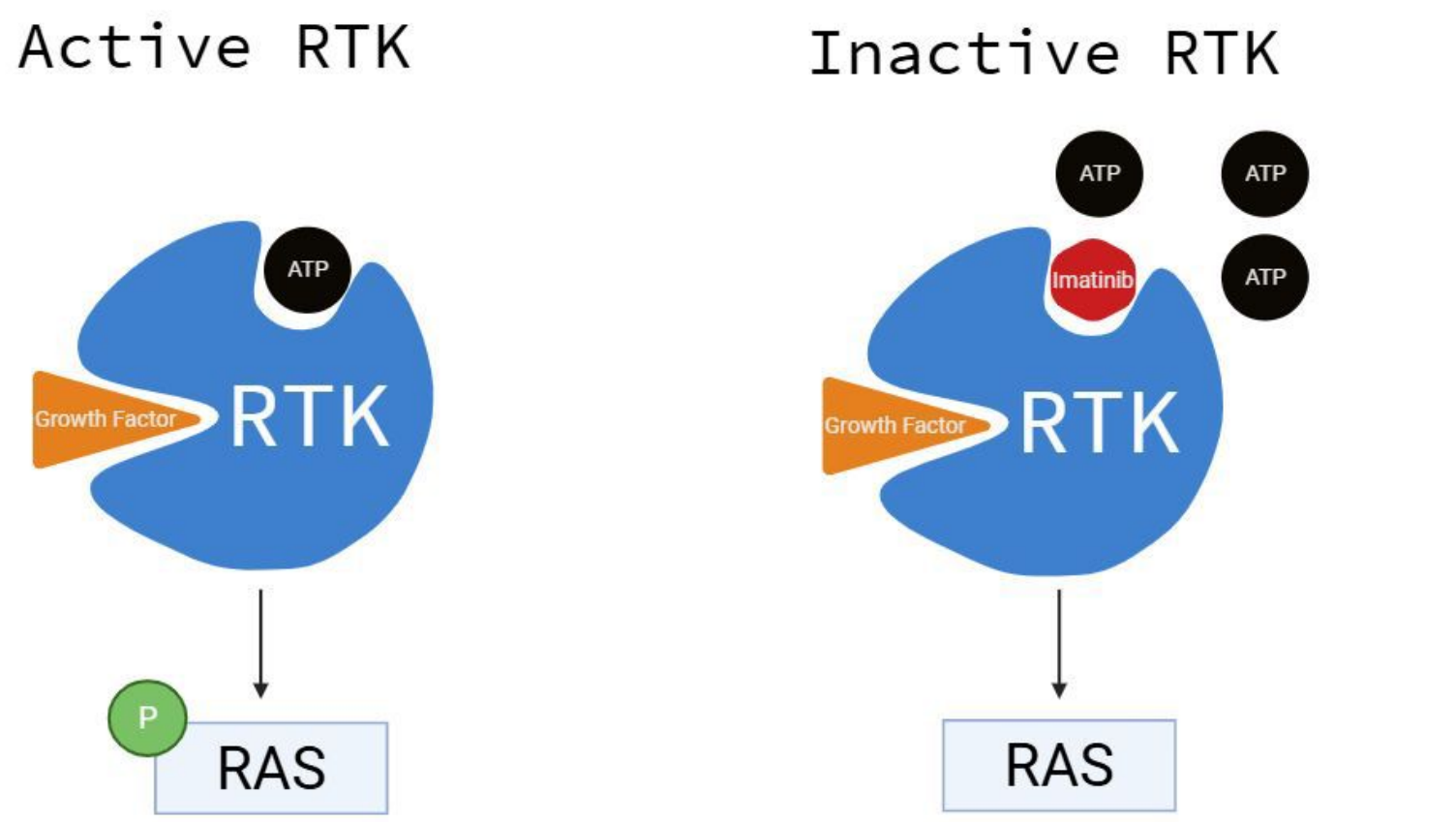


Fig 3 · MAPK pathway with BRAF mutation alongside the AKT pathway. BioRender.

RTK inhibition, a complementary axis



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Fig 4 · Imatinib inhibiting RTK via ATP-competitive binding. BioRender.

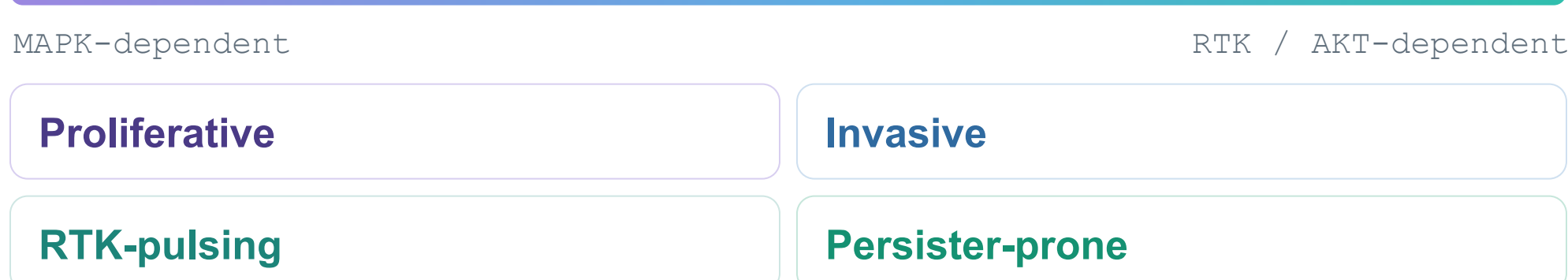
- Persisters depend on the RTK to AKT axis; co-targeting it restores sensitivity (*Villanueva 2010*)
- Imatinib is an ATP-competitive RTK inhibitor that occupies the ATP site (*Klug 2021*)

Drug-combination rationale



- The combination was proposed before, but drug **scheduling and timing** remain understudied (*Villanueva 2010*)

Four virtual patient clusters



- Span the major signaling phenotypes (*Tsoi 2018*; *Verfaillie 2015*), testing phenotype-specific vs. broadly robust schedules

02 Model & Methods

Objectives

- Build a mechanistic ODE model of adaptive resistance coupling MAPK/AKT signaling, PK/PD, and bulk to persister dynamics
- Calibrate** to published data and quantify parameter uncertainty across the plausible range
- Optimize PROTAC + RTKi **schedules** (dose, timing, pulse density, half-life) to control tumor burden
- Test whether optimal regimens are phenotype-specific or broadly robust

Mechanistic model architecture

- MAPK cascade (RTK to RAS to RAF to MEK to ERK) with AKT cross-talk, cFos / CyclinD / DUSP feedback, and bulk to persister switching

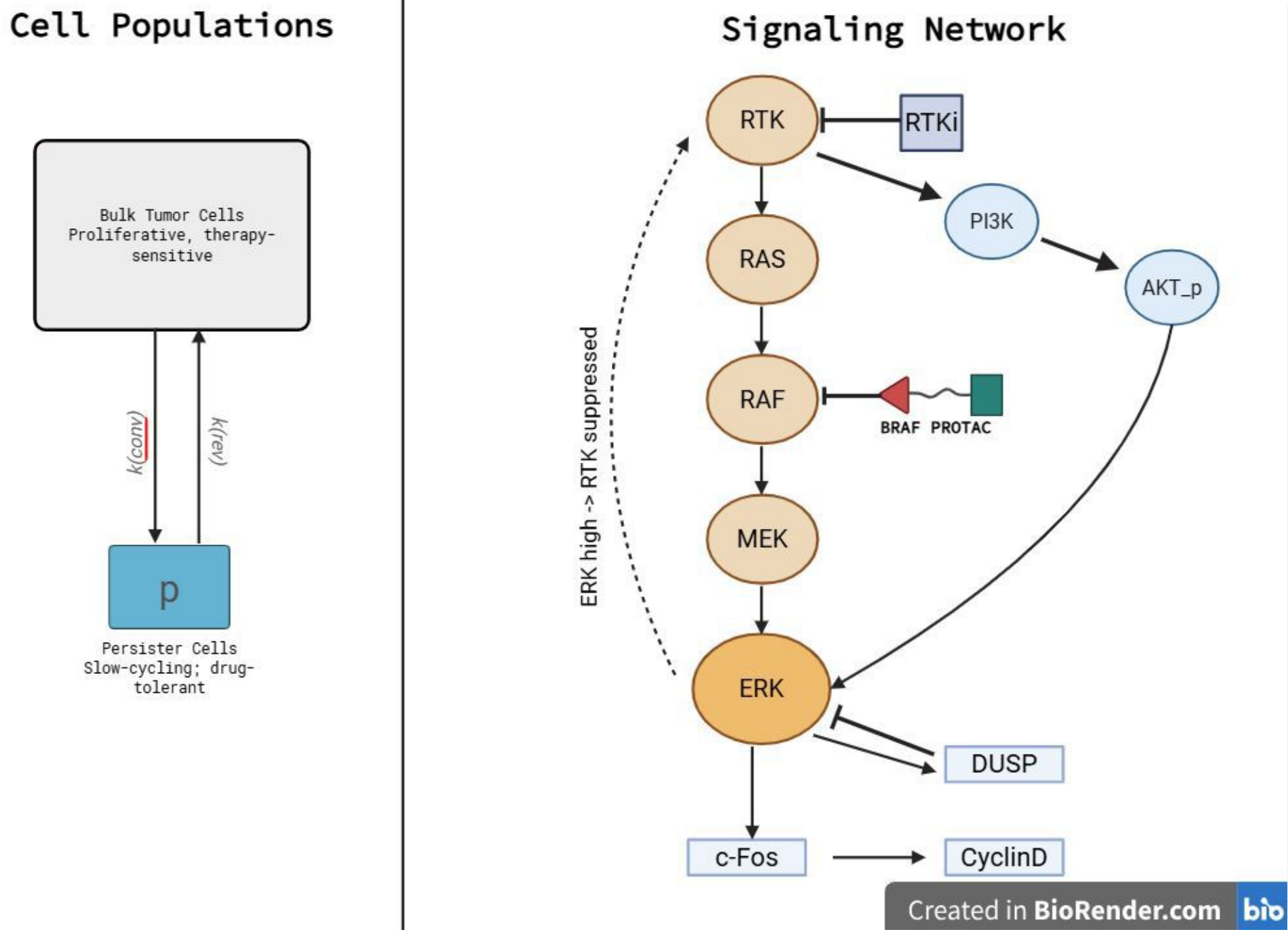


Fig 5 · Mechanistic schematic of the melanoma systems model. BioRender.

Drug exposure & signaling dynamics

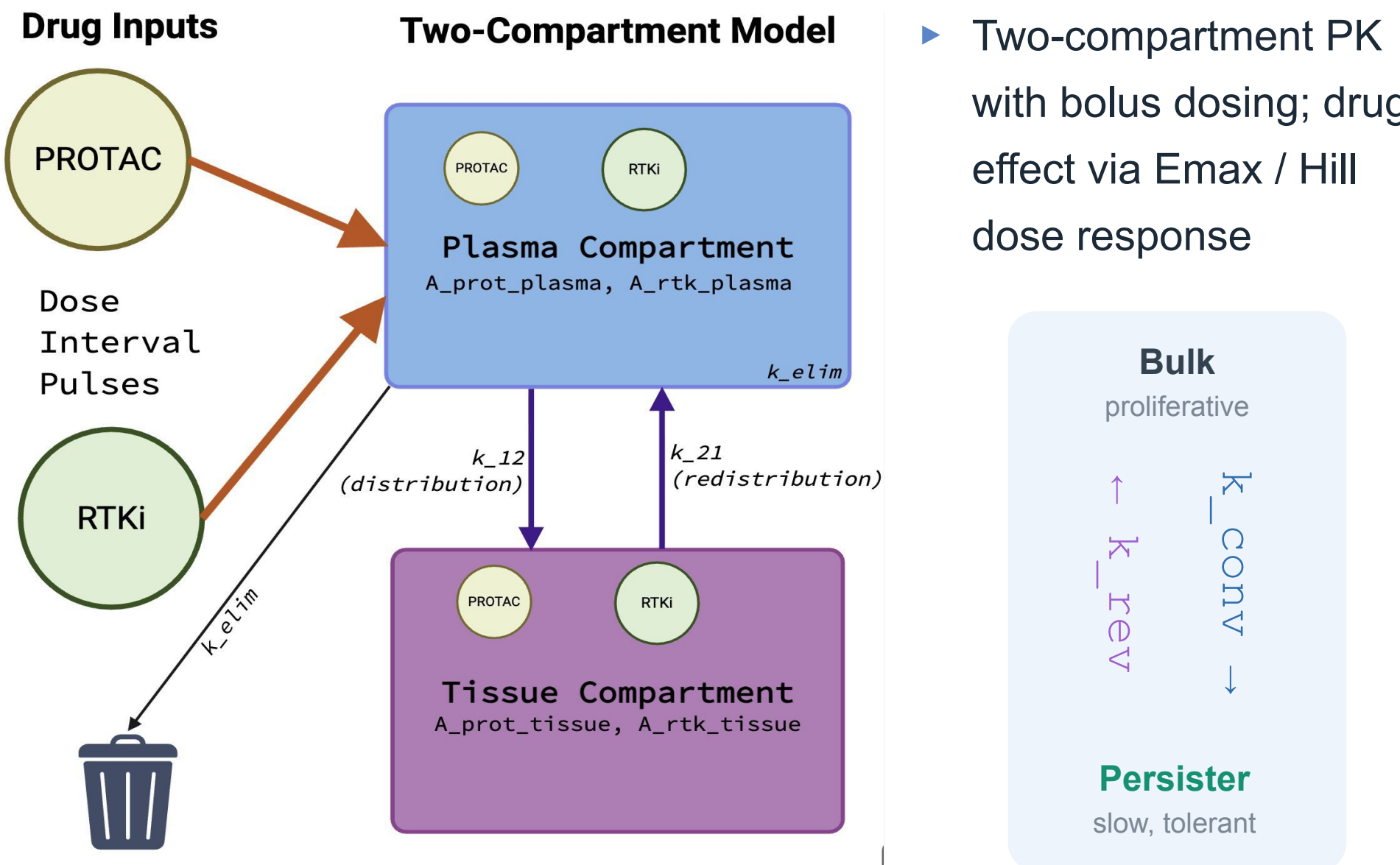


Fig 6 · Two-compartment PK model for PROTAC and RTKi. BioRender.

$$\frac{dMEK}{dt} = k_{mek,act} \cdot RAF \cdot (1 - MEK) - k_{mek,deact} \cdot MEK$$

Fig 7 · Representative ODE for a signaling species (MEK node).

Data calibration (approximate Bayesian computation)

Signaling rebound MAPK drop, rebound & RTK-feedback rise (*Gerosa 2020*)

Single-cell escape 19 to 43% escape sets convert / revert rates (*Yang 2021*)

Dose response Curve shape sets EC50 & Hill coefficient (*Fallahi-Sichani 2013*)

3,000 random sets → score vs. 8 targets → keep best 10% = 300 sets

- Ensemble matches **7 of 8 targets within about 1 SD** (*Toni 2009*)

Parameters are “sloppy”

- Most parameters are only loosely constrained by data, as expected for systems-biology models (*Gutenkunst 2007*)
- Only bulk-cell kill fraction, RTK synthesis, and PROTAC degradation are tightly pinned down
- So every result is reported as a range across all 300 sets, focusing on rank stability**

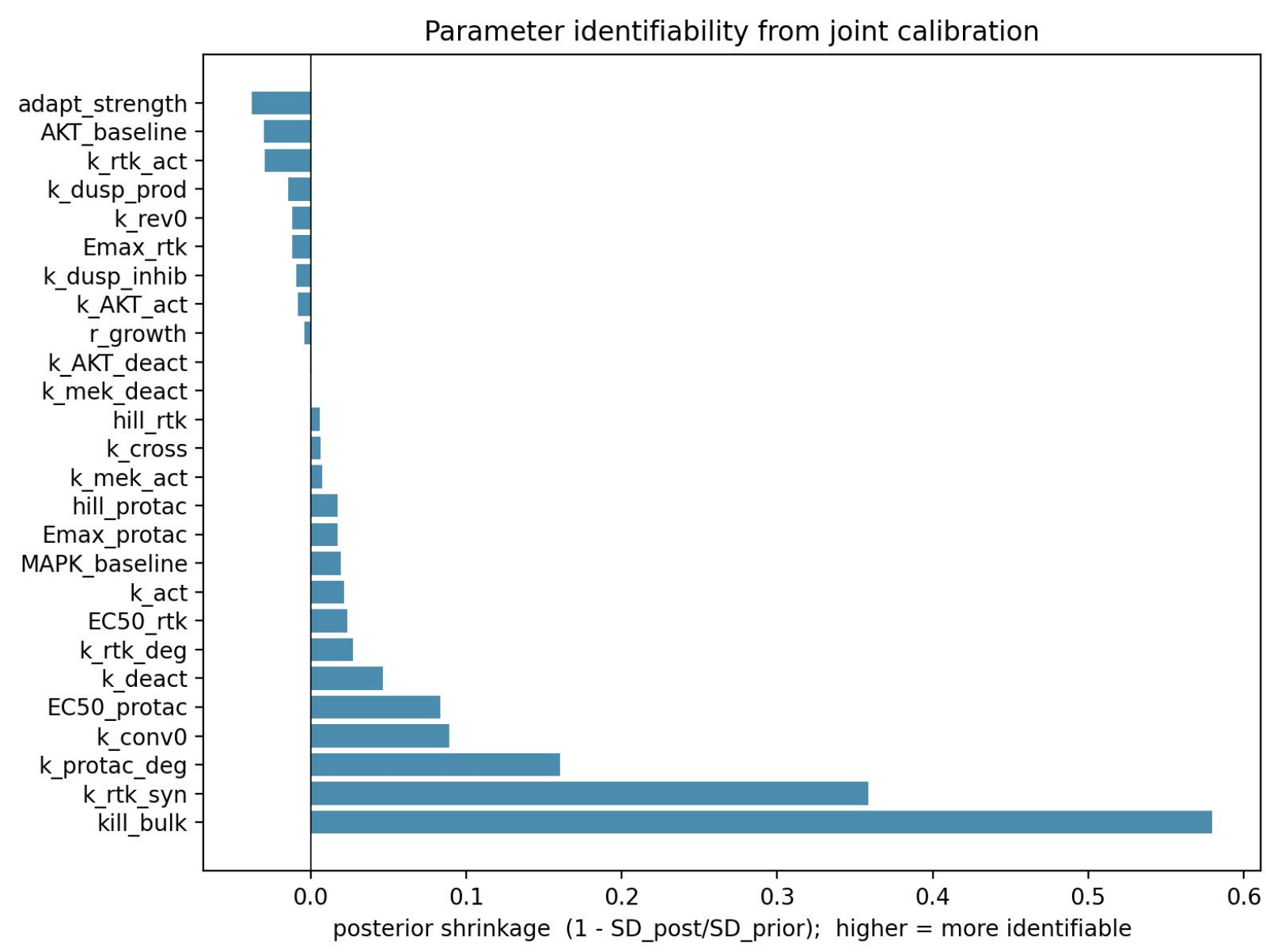


Fig 8 · Posterior shrinkage per parameter after joint fitting. Only kill_bulk, k_rtk_syn and k_protac_deg are tightly constrained.

03 Result

Tumor Control Score (TCS)

- Regimens are scored by **total tumor burden**, the integrated tumor size over the treatment course
- Measured over a **20-day window**, long enough to separate durable control from regrowth

Model reproduces adaptive resistance

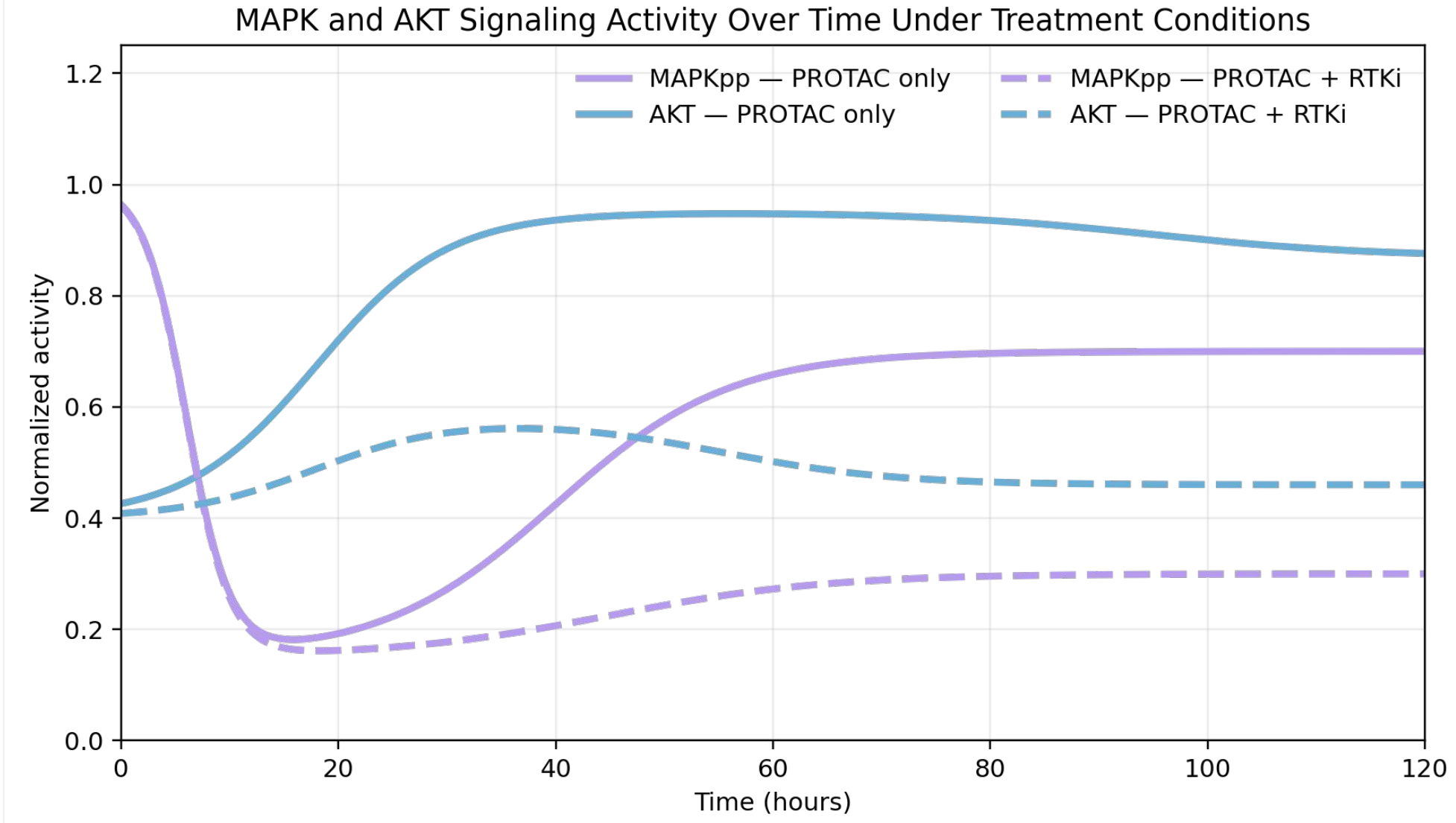


Fig 9 · MAPK and AKT activity over time: monotherapy rebounds; combination sustains suppression.

- Under PROTAC alone, MAPK drops then rebounds as AKT climbs; the combination holds both down

Combination beats either single drug

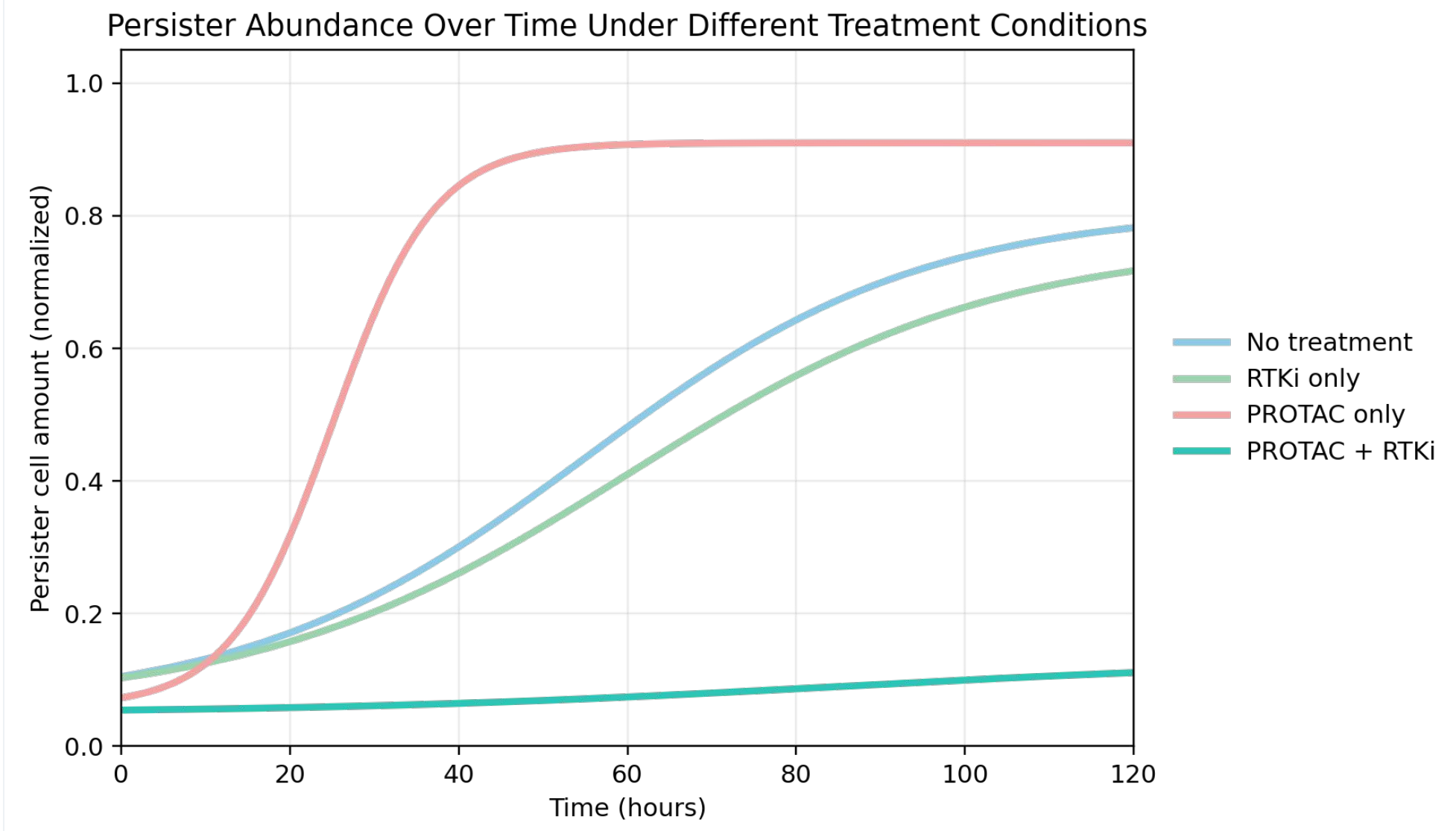


Fig 10 · Persister abundance under no treatment, RTKi only, PROTAC only, and combination.

- Ordering **combo > PROTAC > RTKi > none** held across all 300 calibrated sets; hitting both pathways removes the escape route

Scheduling robustly controls burden

- 300+ runs per tumor type, reported as means with 95% confidence bands

- Aggressive, sustained schedules hold final tumor to **15 to 22%** of max; minimal or single-drug schedules regrow to **70 to 80%**

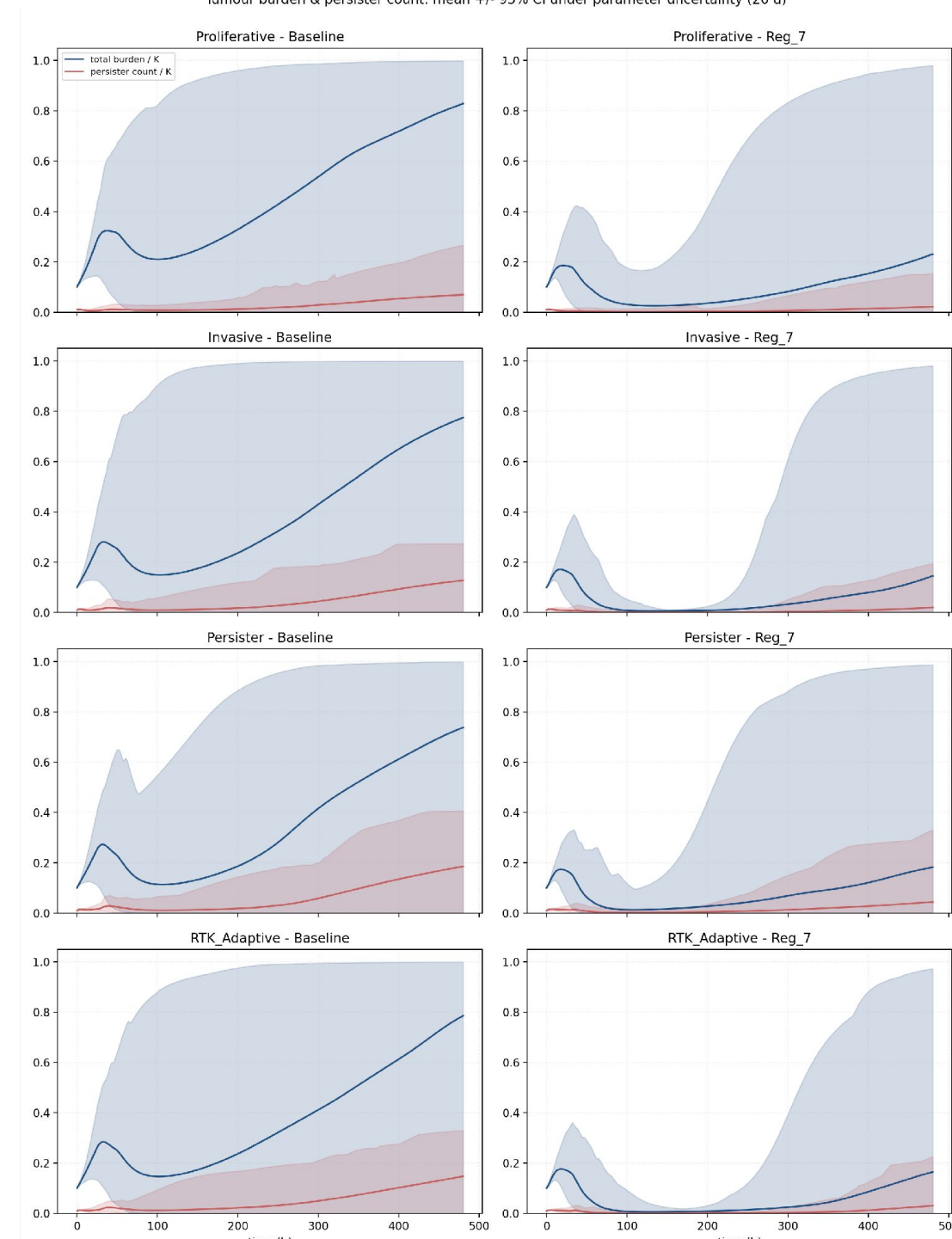


Fig 11 · Burden and persister count over 20 days, mean and 95% CI across 300 sets. Reg 7 holds burden low with tight bands; baseline regrows.

Burden separates control from regrowth

- Across schedules, total tumor size spans about 10³-fold over the 20-day window
- Ranking on burden cleanly orders schedules from durable control to full regrowth

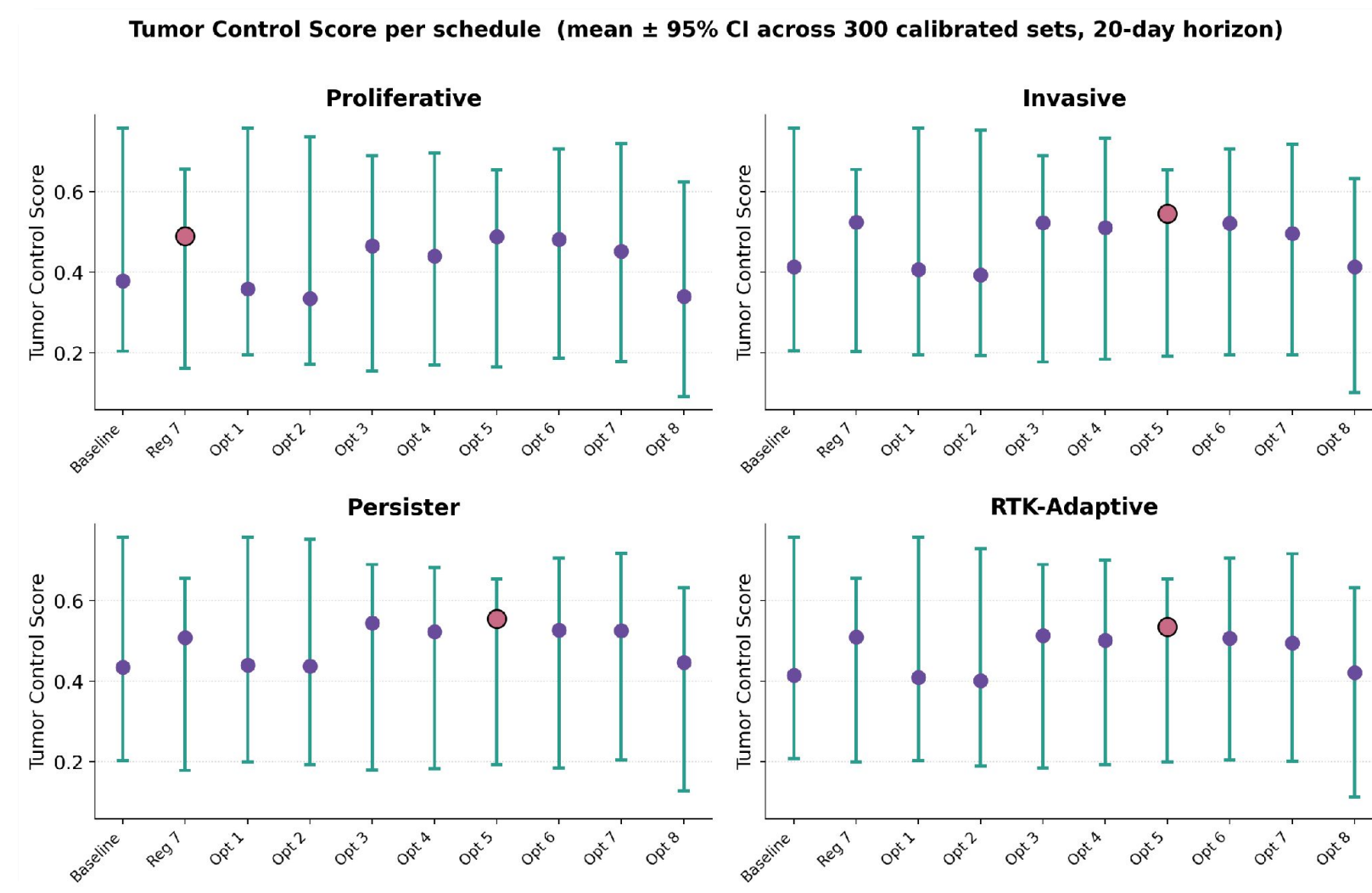


Fig 12 · TCS per schedule per tumor type, mean and 95% CI across 300 sets. Highlighted point is the best schedule per type.

04 Findings & Conclusions

A winning family, not one regimen

- Across 123 tested schedules, every top performer shares **frequent high-dose PROTAC pulses plus steady, long-lasting RTK inhibition**
- The winner is a **robust family** of schedules rather than a single fragile recipe

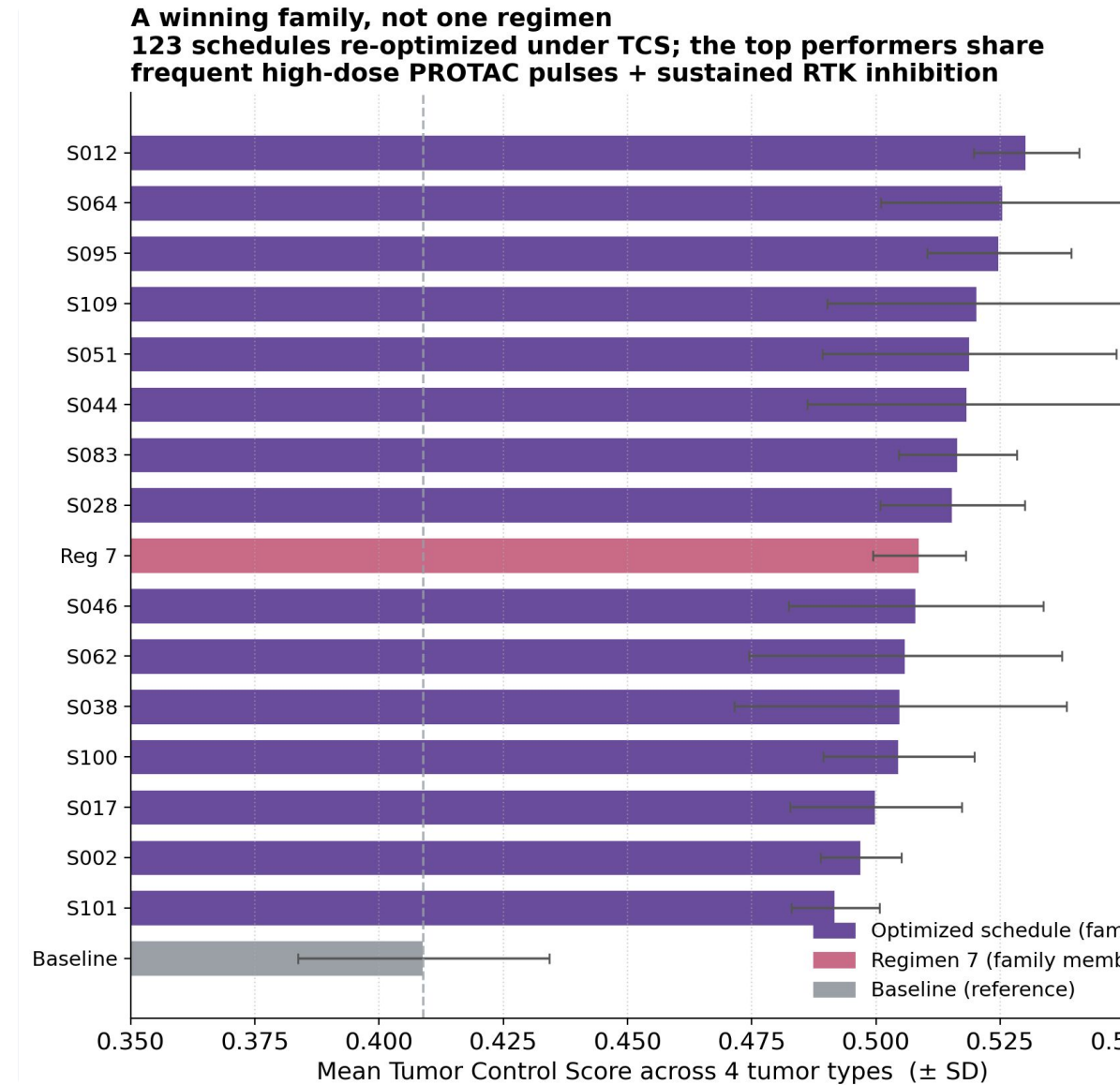


Fig 13 · Top optimized schedules and Reg 7 form a tight family (TCS about 0.49 to 0.53), all above baseline (dashed, about 0.41).

Regimen 7 sustains suppression

MAPKpp — Baseline (combo) MAPKpp — Regimen 7

AKT — Baseline (combo) AKT — Regimen 7

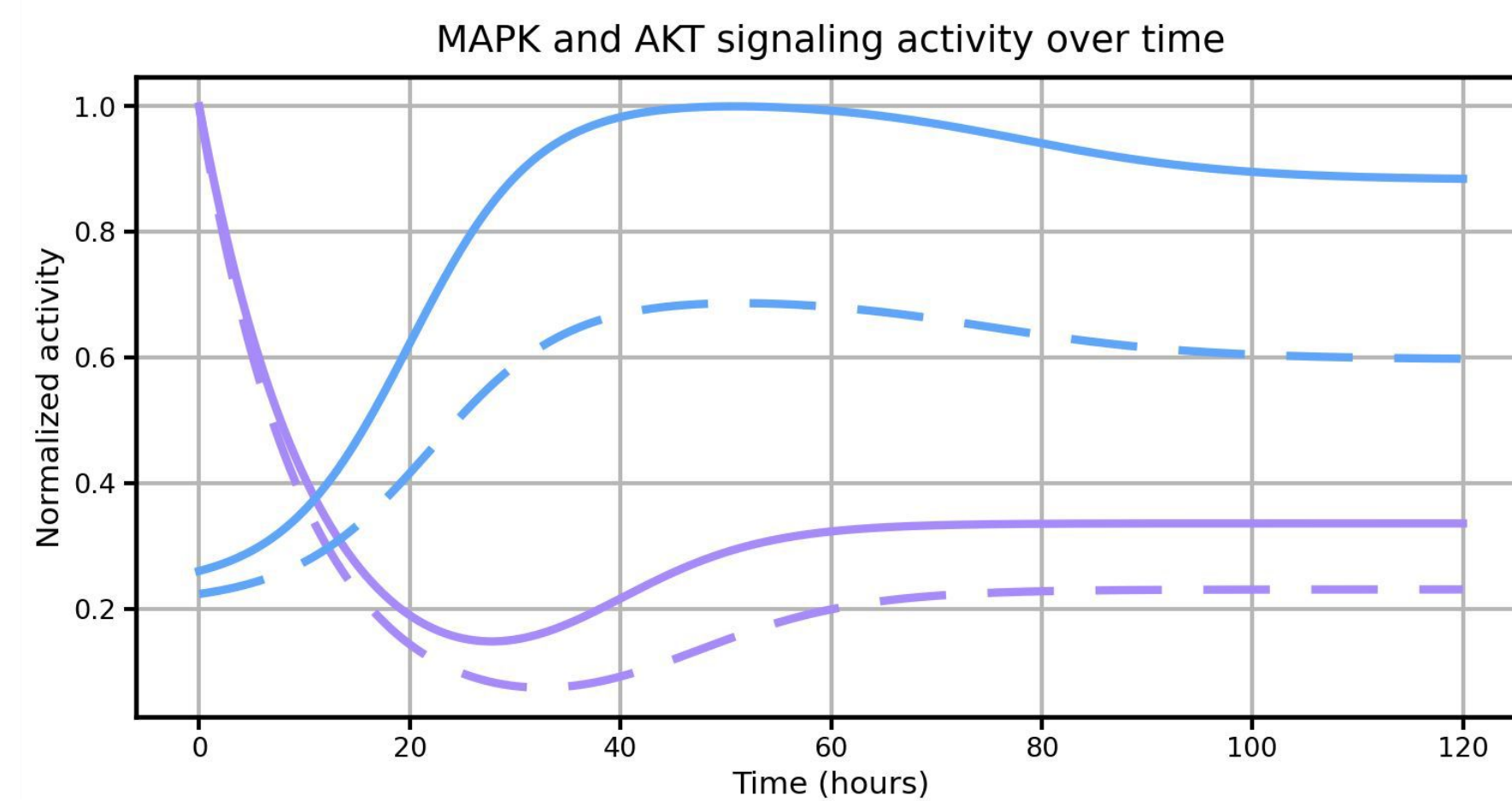


Fig 14 · MAPK and AKT activity over time: baseline combination vs. Regimen 7.

- A longer RTKi half-life plus denser PROTAC pulsing keeps MAPK/AKT and persists below baseline combination therapy

Robust across scoring choices

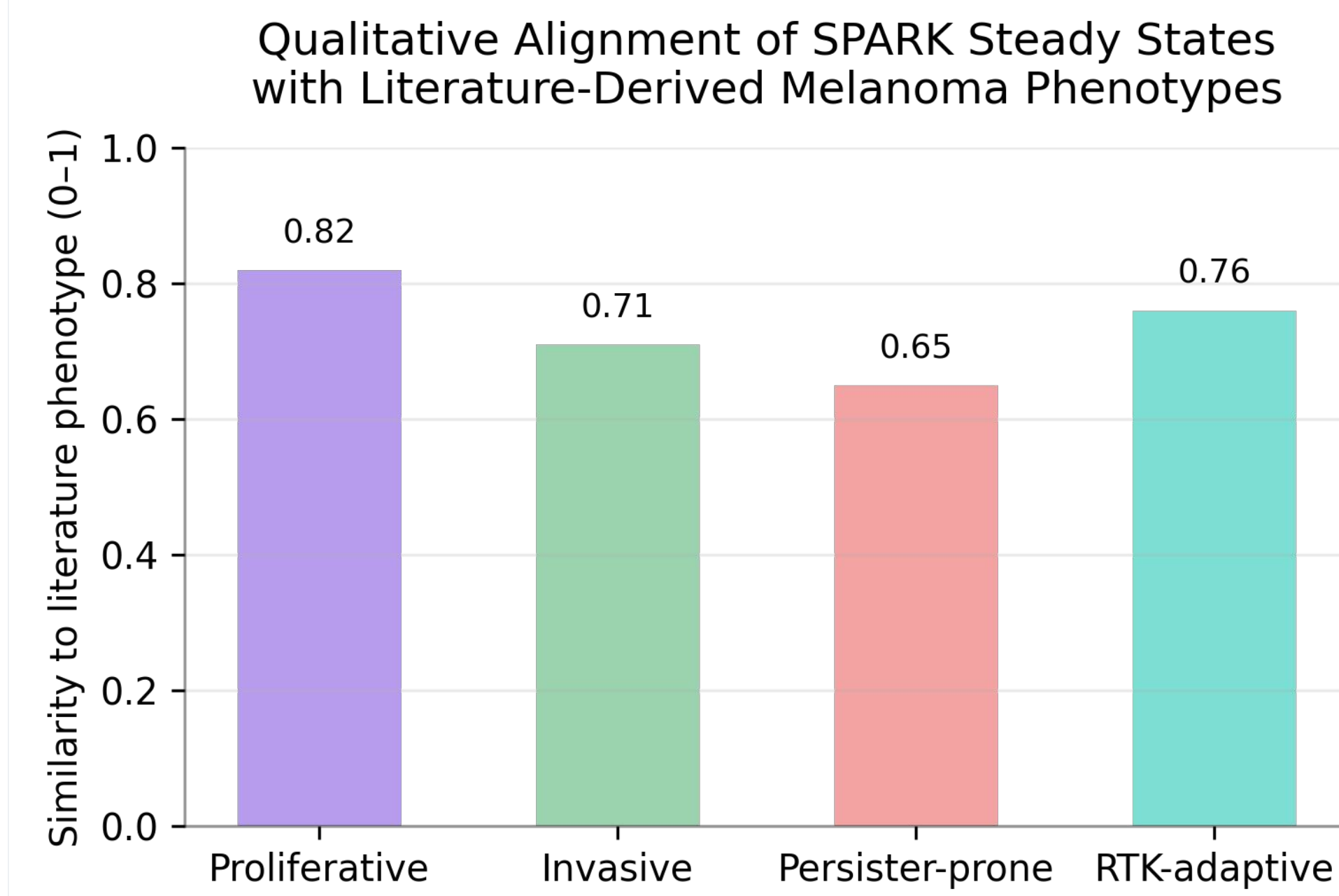


Fig 15 · Simulated steady states align with literature-derived melanoma phenotypes.

- Rankings barely shift under re-weighting (rank correlation about 0.94); every efficacy-focused weighting favors the same family
- Steady-state signaling matches literature phenotypes (average similarity about 0.74)

Discussion & limitations

- Timing, not the drugs**, drives the difference: a robust, underused lever confirmed across the full calibrated ensemble
- Response is partly shared across clusters, so grouping by adaptive behavior may suffice
- Limits:** loosely-constrained parameters (reported as ranges); suppression depth about 26% vs. 9% reference; 2-state, deterministic, well-mixed model

Conclusions

- In a data-calibrated, uncertainty-aware model, **treatment scheduling is a robust lever** against adaptive resistance
- A family of **high-PROTAC plus sustained-RTK** schedules controls tumor burden best across tumor types and parameter uncertainty
- Supports a shift from a fixed, mutation-centered view toward control of dynamic, reversible cell states

Selected references

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